

Handling Chemicals in the EKL laboratories

July 2023 - Standard EKL version

Survey

Who had additional chemical training after middle school?

Who of you has handled (dangerous) chemicals in a lab?

Introduction

During this course you will get instructions on how you can safely work with chemicals.

You must always be aware of working safely:

- Think of your own safety, the safety of other users and the laboratory.
- You must always keep safety in the back of your mind while working in the labs.

It takes time to get experience in safely using chemicals: **Practice Makes Perfect!**

After this **Handling with chemicals in the EKL laboratories** presentation:

- You must apply in NIS for a hands-on training for the specific location where you will perform your experiments.

Chemicals at EKL

You may be working with chemicals yourself, or you can be in close range to locations where chemicals will be used by others.

- All chemicals have a MSDS (Material Safety Data Sheet).
- Most of these chemicals are very hazardous.

Chemicals used at EKL:

- There are $\pm$ 300 different liquid and solid chemicals present.
- More than 35 different gasses are present. These are mostly being used in equipment.

It is important to know:

- How dangerous these chemicals are.
- How to use the personal protective equipment (fume hood / wetbench, safety glasses, apron, etc.).
- How to handle in case of incidents or accidents.

Safety & Introduction Course repetition points

Important:

- While transporting, pouring and mixing chemicals: **Your buddy **must be** in the same area.**
- While working with chemicals: **You need a buddy.**

Before starting your training in the cleanroom, make sure that your trainer makes you familiar with the locations of the following items:

- Escape routes to the emergency exits.
- Fire extinguishers and fire alarm button.
- Emergency button and calamity button.
- Diphoterine sprays and Diphoterine suitcase (emergency rinsing solution for splashes of chemical products).
- Hexafluorine suitcase (first aid decontamination agent for HF).
- Eye shower bottles.
- Water showers

Use of chemicals during lab opening hours

➤ **Office hours:** CR + MEMS ⇒ Monday - Friday 8:00 - 17:00, except holidays

It is **ALLOWED** to transport, pour and mix (approved) chemicals in this time period.

- For working in the CR wet benches: A buddy is required.
(these bathes are already prepared)
- While **handling** chemicals: Your buddy must be **in the same area**.
(handling includes transport, pouring and/or mixing)

➤ **Extended opening hours:** CR ⇒ Monday - Friday 17:00 - 22:00, weekends and holidays
MEMS ⇒ Monday - Friday 17:00 - 20:00, weekends and holidays

It is **NOT ALLOWED** to transport, pour and mix chemicals, also **NOT** when a buddy is present.

- For working in the CR wet benches: A buddy is required.
(these bathes are already prepared)

☺ Exceptions for these rules can only be obtained from the **Safety Officer**.

☹ Violation of the above rules may result in **revoked access rights**.

What to do before starting an experiment involving chemicals

If you do not have experience in working with chemicals:

You need permission, training and help! If you are unsure: ASK!

Create a description of the steps involved in your experiment.

- Talk with your supervisor, mentor, fellow students or staff members and perform a literature study.
- Collect all the information.
- Create a flowchart for the experiments incorporating information about the used chemicals and equipment.
- Create a Risk Assessment document and submit this to the safety officer.

Check with the wet bench team if the chemicals that you need to use are present in EKL.

- If yes: It is allowed to use them.
- If no : You need to ask for approval to introduce them from the safety officer and the process manager.
(See slide: [New chemicals in EKL](#)).

Additional information

Collect as much information about the used chemicals as possible, like:

- Amount.
- Concentrations.
- Density.
- Purity.
- Mixing capability with other chemicals.
- The applied temperature.
- Etch rates and selectivity.

Make a list of all additional needed materials:

- Tissues.
- Glass ware.
- Heaters.
- Etc..

If you want to heat up organic chemicals: Always use an Au Bain Marie setup.

A Flowchart is needed for processing in CR100

It is prohibited to start a process without an **updated and approved flowchart.**

- A flowchart contains a detailed description of all process steps, including the process parameters: (e.g. gas flows, power, temperature, concentration, time, pressure etc.).

Contact the EKL process manager when you are creating your flowchart:

- Acquire a run number, fill in the flowchart front page template and mail it.

Do you use wafers, chemicals or materials that are not CMOS compatible?

- Then your flowchart must comply with the EKL PCC document and process guide table, and needs approval from the contamination officer.

Are you performing a small test or experiment?

- You also need an **updated approved** flowchart with a run number (p-number).

Are you making significant changes in your flowchart?

- **Then you need a new updated approved flowchart with a new run number.**

Example: Chemicals involved in TMAH etch of a Si cavity

1. Cleaning wafers (HNO_3 99% + HNO_3 69% 110°C) .1
2. Deposition masking layer.
3. Depositing photo sensitive resist layer.
 - HMDS. 2
 - Photo Resist. 3
4. Exposure.
5. Development with TMAH ca 0.25M. 4
6. Etching masking layer (Wet or RIE). 5
7. Removing photo resist (Acetone or Oxide plasma). 6
8. Cleaning wafers (HNO_3 99% + HNO_3 69% 110°C) .7
9. (B)HF dip. 8
10. TMAH etch. 9
11. Cleaning SAL (HNO_3 99%). 10
12. Cleaning CR100 (HNO_3 99% + HNO_3 69% 110°C). 11
13. BHF removing masking layer. 12
14. Cleaning (HNO_3 99% + HNO_3 69% 110°C) .13

What did we learn from this TMAH etching process

This process block for wet Si etching contains:

- **13** steps with 1 or more liquid chemicals.

It is possible to make mistakes and cause an incident or accident.

- It is important to know how to act when this happens.

It is important to know how chemicals can harm **you and/or **your surrounding area**.**

- All this information can be found in a MSDS (Material Safety Data Sheet).

Where can you find a MSDS?

- Check the Internet.
- Check the holders that are mounted on the storage cabinets.
A MSDS for each chemical in a cabinet is archived in that holder.
- In books with a collection of all these sheets. Ask the Wetbench team.

Brief introduction to chemical knowledge

We can distinguish 2 categories of chemicals:

- Organic chemicals.
- Inorganic chemicals.

What are organic chemicals:

- An organic compound is a chemical that contains **carbon-hydrogen** or **carbon-carbon** bonds.

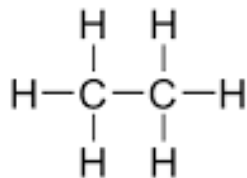
What are inorganics chemicals:

- All other chemicals.

Organic chemicals

Types of organic chemicals:

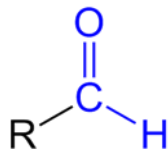
- Alkanes



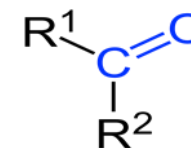
- Alcohols



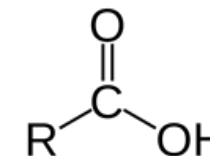
- Aldehydes



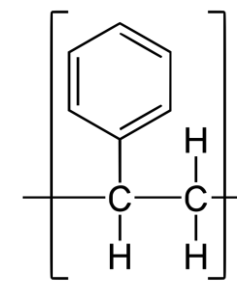
- Ketones



- Carboxylic acids



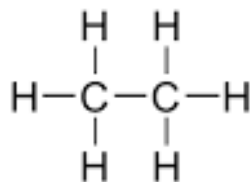
- Polymers



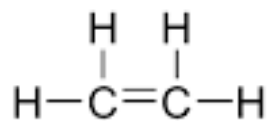
Organic chemicals

Types of organic chemicals:

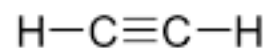
- Alkanes



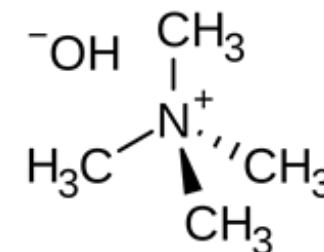
- Alkenes



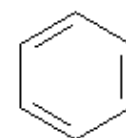
- Alkynes



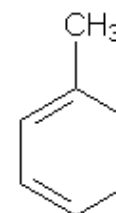
- Organic base
(a part is inorganic)



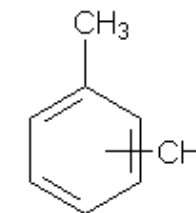
- Aromatic substances



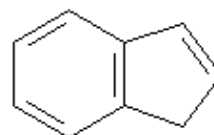
benzene



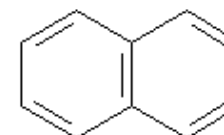
toluene



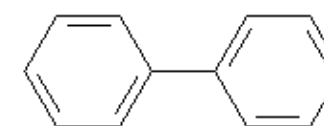
o-, *m*-, *p*-xylene



indene



naphthalene



biphenyl

Inorganic chemicals

Types of inorganic chemicals:

- **Pure elements:**

- Solids : Al, Mo, Zr, Sb, I₂
- Liquids : Hg, Br₂
- Gases : N₂, H₂, Cl₂

- **Chemical molecules:**

- Solids : Al₂O₃, SiO₂, Si_xN_y
- Liquids : Br₂, H₂O
- Gases : Cl₂H₂Si, GeH₄, SF₆

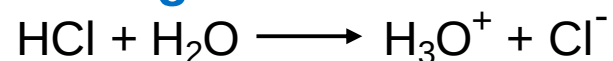
- **Salts** : NaCl, NH₄Cl, Na₂S, KClO₄

- **Inorganic acids** : HCl, HNO₃, H₂SO₄

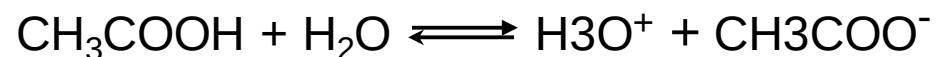
- **Inorganic bases** : KOH, NH₄OH

Weak and strong acids/bases.

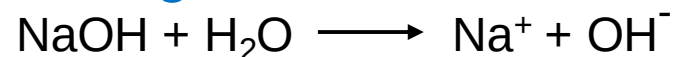
- **Strong acids**



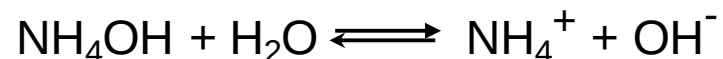
- **Weak acid**



- **Strong base**



- **Weak base**



Strong	Weak
Strong acids or bases are completely dissociated when dissolved in water.	Weak acids or bases are partially dissociated when dissolved in water.
The dissociation constant has a High value .	The dissociation constant has a Low value .
All H or OH is released as an ion into the solution.	Not all H or OH is released as an ion into the solution.

Examples of strong acids/bases

Examples of strong acids:

Formula	Name
■ HClO_4	Perchloric acid
■ HCl	Hydrochloric acid
■ HBr	Hydrobromic acid
■ HI	Hydroiodic acid
■ HNO_3	Nitric acid
■ H_2SO_4	Sulfuric acid

Examples of strong bases:

Formula	Name
■ LiOH	Lithium hydroxide
■ NaOH	Sodium hydroxide
■ KOH	Potassium hydroxide
■ TMAH	Tetra methyl ammonium hydroxide

Examples of weak acids/bases

Examples of weak acids:

Formula	Name
▪ HF	Hydrofluoric acid
▪ HNO ₂	Nitrous acid
▪ H ₂ S	Hydro sulfuric acid
▪ CH ₃ COOH	Acetic acid
▪ H ₃ AsO ₄	Arsenic acid
▪ BF ₃	Boor fluorine

Examples of weak bases:

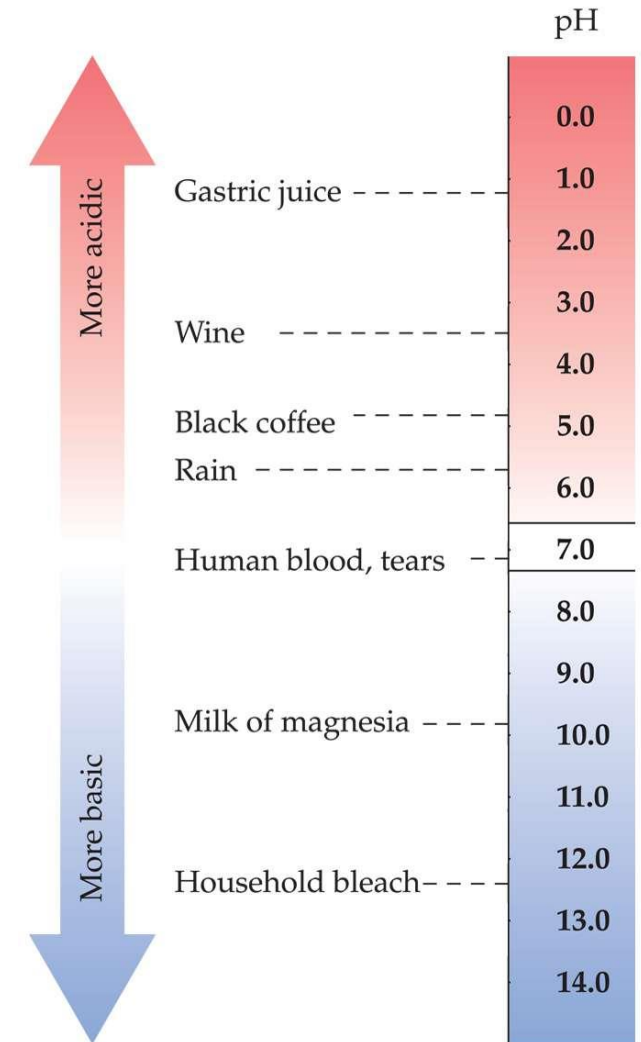
Formula	Name
▪ NH ₄ OH	Ammonium hydroxide
▪ Al(OH) ₃	Aluminum hydroxide
▪ NaHCO ₃	Sodium Bicarbonate
▪ Mg(OH) ₂	Magnesium Hydroxide)
▪ CH ₃ NH ₂	Methylamine

pH variation

pH Definition:

$$\text{pH} = -\log [\text{H}_3\text{O}^+] \quad \text{H}_3\text{O}^+ = \text{acid part}$$

The most acids/bases in EKL are highly concentrated:



Section 7.
How to handle the chemical and how to store it

Section 8.
Personal Protection: Gloves, Goggles, Apron, Fume Hood

Exposure limits:
PEL= permissible exposure limit
TWA= time-weighted average (8 hrs)
STEL= short-term exposure limits (15-30 min)

Section 9.
Look at Boiling point, Vapor Density (heavier than air), Odor threshold. If this threshold is higher than the TWA/STEL than be very careful!
The vapor pressure (in kPa)*10⁴ gives the concentration in ppm. Compare this with the STEL value to see how much fresh air you need. Here: VP: 240000pm STEL: 2400 ppm, so you need at least 100x dilution with fresh air.

Section 10.
Describes the conditions or reactions to be avoided. Also provides some indication about anticipated shelf life.

Section 11.
More detail on how the material may injure you. Acute (short exposure) and chronic (long-term) effects are listed along with their target organs.
Oral (ORL), inhalation (IHL), and skin absorption (SKN) toxicity data on test animals can be included.

[illegible]

Section 12.
Ecological impact if large amounts (e.g., tank car) of the chemical spill near a river or lake.

Section 13.
Suggested disposal methods for laboratory quantities of chemicals.

Section 14.
Department of Transportation shipping information is included for your school district, emergency responders, and transport/shipping departments.

Section 15.
Regulatory information used by regulatory compliance personnel.

Section 16.

MSDS for Nitric acid

Sections in the Material Safety Data Sheet:

- Chemical name and formula.
- Cass number.
- Physical properties.
- Important characteristics.
- Hazards / Symptoms.
- Preventive measures.
- Fire extinguishing / First aid.
- Spillage (how to clean)
- Storage.
- Labeling:
 - Hazard labels.
 - NFPA: Hazard diamond.
 - Risk and Safety (R&S) phrases.
- Remarks / Notes.
- Handling in case of emergency / Influence on environment.

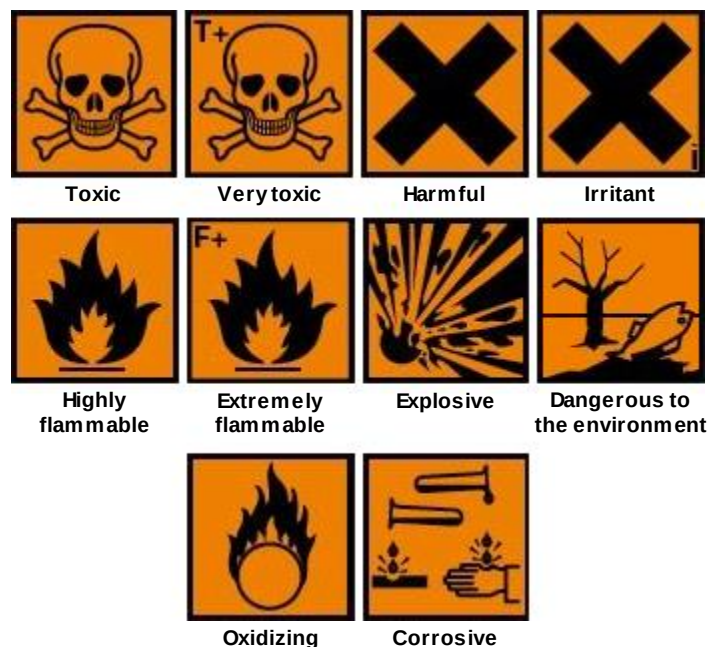
CAS-No: [7697-37-2]		HNO3	
NITRIC ACID (above 70%)			
PHYSICAL PROPERTIES		IMPORTANT CHARACTERISTICS	
Boiling point, °C 83 (100%) Melting point, °C -42 (100%) Relative density (water = 1) 1.5 Relative vapor density (air = 1) 2.2 Relative density at 20 °C of saturated mixture vapor/air (air = 1) 1.06 Vapor pressure, mm Hg at 20 °C 43 (100%) Solubility in water ∞ Relative molecular mass 63.0		COLORLESS TO RED AQUEOUS SOLUTION OF NITRIC ACID IN VARIOUS CONCENTRATIONS ABOVE 70% WITH PUNGENT ODOR Vapor mixes readily with air. Decomposes when heated or exposed to light, giving off toxic gas (→ nitrous vapors). Strong oxidant which reacts violently with combustible substances and reducing agents. Strong acid which reacts violently with bases and is corrosive. Reacts violently with finely dispersed metals, organic solvents and many other substances, with risk of fire and explosion. TLV-TWA 2 ppm 5.2 mg/m³ STEL 4 ppm 10 mg/m³	
Gross formula: HNO3		Absorption route: Can enter the body by inhalation. Harmful atmospheric concentrations can build up very rapidly on evaporation at 20 °C. Immediate effects: Corrosive to the eyes, skin and respiratory tract. Inhalation of vapor/fumes can cause severe breathing difficulties (lung edema). In serious cases risk of death.	
HAZARDS/SYMPTOMS		PREVENTIVE MEASURES	FIRE EXTINGUISHING/FIRST AID
Fire: non-combustible, but makes other substances more combustible; many chemical reactions can cause fire and explosion.	avoid contact with combustible substances, solvents and many other substances.	in case of fire in immediate vicinity: use any extinguishing agent.	
Explosion: many chemical reactions can cause explosion.		in case of fire: keep tanks/drums cool by spraying with water.	
	STRICT HYGIENE		
Inhalation: corrosive; sore throat, cough, shortness of breath, severe breathing difficulties.	ventilation, local exhaust or respiratory protection.	fresh air, rest, place in half-sitting position, take to hospital.	
Skin: corrosive; redness, pain, burns.	gloves, protective clothing.	remove contaminated clothing, flush skin with water or shower, call a doctor.	
Eyes: corrosive; redness, pain, impaired vision.	face shield, or combined eye and respiratory protection.	flush with water, take to a doctor.	
Ingestion: corrosive; sore throat, abdominal pain.		rinse mouth, immediately take to hospital.	
SPILLAGE		STORAGE	LABELING / NFPA
collect leakage in sealable containers, soak up with sand or other inert absorbent and remove to safe place, flush away remainder with water, do not use sawdust or other combustible absorbents; (additional individual protection: CHEMICAL SUIT).	keep in cool, dark place, separate from combustible substances, reducing agents and bases; ventilate at floor level.	R: 8-35 S: 23-26-36 Note B  Oxidant  Corrosive	
			
NOTES			
In concentrations above 80% is referred to as fuming nitric acid. Physical properties of 100% nitric acid are given. Flush contaminated clothing with water (fire hazard). UN No. of red fuming nitric acid: 2032. Lung edema symptoms usually develop several hours later and are aggravated by physical exertion; rest and hospitalization essential. As first aid, a doctor or authorized person should consider administering a corticosteroid spray. UNDER NO CIRCUMSTANCES add water to acid; when diluting ALWAYS add acid to water. Unbreakable packaging preferred; if breakable, keep in unbreakable container. Packaging: dark glass, aluminum, stainless steel or suitable plastic.			
Transport Emergency Card TEC(R)-9A		HI: 885; UN-number: 2031	

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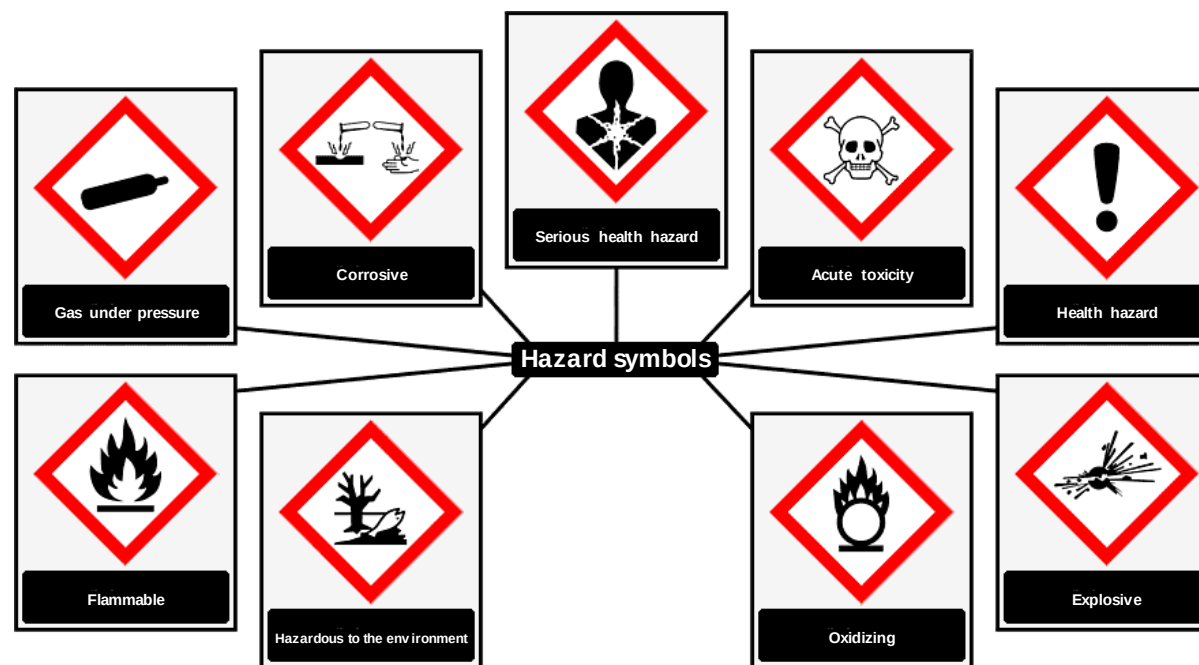
Labelling: Hazard symbols

The following hazard symbols can be found on bottles containing chemicals, and on some Material Safety Data Sheets.

Old hazard symbols
that were used until 2016.



New GHS (Globally Harmonized System)
symbols, in use since 2016.



Pictograms: Health hazard

GHS Pictogram	Hazards	Intake	H phrase
	- Corrosive, - Irritant effect on skin, - Severe eye irritation.	Contact.	Causes severe skin burns and eye damage.
	Acute toxicity.	- Oral, - Dermal, - Inhalation.	- Causes severe eye damage. - Danger to life if swallowed, in contact with skin and if inhaled. - Toxic if swallowed, in contact with skin and if inhaled.
	- Acute toxicity, - Skin corrosion/irritation, - Severe eye irritation/damage, - Skin sensitization, - Specific target organ toxicity.	- Oral, - Dermal, - Inhalation.	- Harmful by ingestion, skin contact and inhalation. - Causes skin irritation and severe skin irritation. - May cause allergic skin reactions and respiratory irritation. - May cause drowsiness and dizziness.
	- Respiratory sensitization, - Germ cell mutagenicity, - Carcinogenicity, - Reproductive toxicity, - Specific target organ toxicity from single or repeated exposure, - Aspiration hazard.	- Oral, - Dermal, - Inhalation.	- May cause allergy, asthma-like symptoms, or breathing difficulties if inhaled. - May (probably) cause genetic defects; may (probably) cause cancer. - May (probably) impair fertility or harm the unborn child. - May harm infants through breast milk; damages organs. - May damage organs (with prolonged or repeated exposure). - May be fatal if swallowed and enters airway.

Pictograms: Fire and explosion hazard

GHS pictogram	Danger	H phrase
	- Explosive substances, - Mixtures and articles containing explosives.	- Instable, explosive, - Danger of mass explosion (in case of fire), - Great danger from splinters, explosive and throwing pieces, - Danger from fire, air pressure and splinters, explosive and throwing pieces, - Heating can cause explosion and/or fire.
	- Flammable gases, aerosols, liquids and/or solids, - Self-reactive substances and mixtures, - Pyrophoric liquids and/or solids, - Substances and mixtures capable of self-heating, - Substances and mixtures which, in contact with water, emit flammable gases, - Organic peroxides.	- Extremely flammable gas, - Flammable gas and/or aerosol, - Liquid and vapor (slightly/extremely) flammable, - Flammable solid; heating may cause fire and/or explosion, - Self-ignites in contact with air, - Self-heating (in large quantities), - May catch fire; contact with water produces flammable gases (which May ignite spontaneously).
	- Oxidizing gases, - Liquids, - Solids.	- Oxidizing agent, - May cause and/or intensify fire or explosion.

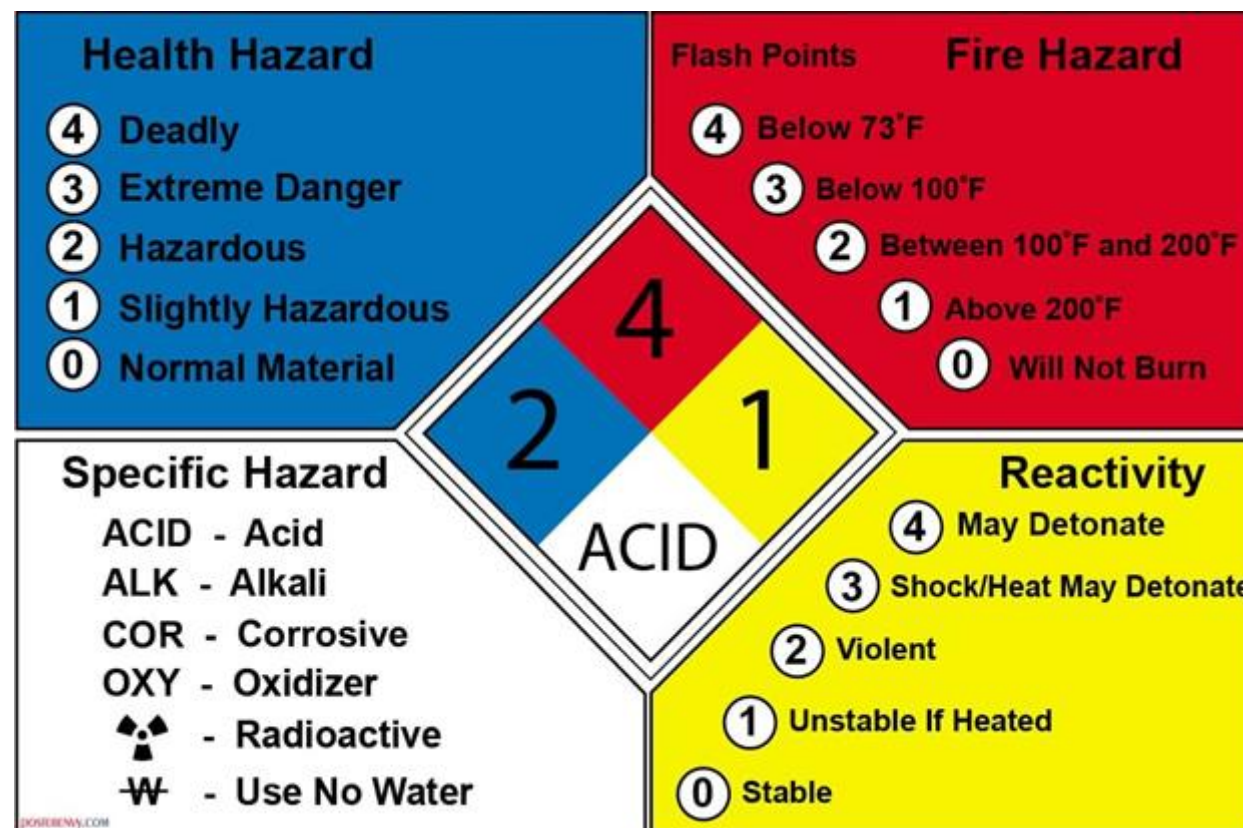
Pictograms: Physical, chemical and environmental

GHS pictogram	Danger	H phrase
	• Gases under pressure, • Compressed, liquefied, dissolved, • Cryogenic liquefied gas.	• Contains gas under pressure, • May explode if heated, • Contains cryogenic gas, • May cause cold burn or injury
	• Corrosive, • Corrosive to metals	• Corrosive, • May be corrosive to metals
	• Acutely hazardous to the aquatic environment, • Chronically hazardous to the aquatic environment, • Harmful to the ozone layer.	• (Very) toxic to aquatic organisms; • Possible long-term effects, • Harmful to aquatic organisms (long-term effects), • Harmful to the ozone layer.

Labelling: NFPA diamond

The NFPA (National Fire Protection Agency) diamond shows a quick indication of the hazards and the reactivity of a chemical.

- There are 4 fields:
 - Left (blue) = Health Hazard
 - Top (red) = Fire Hazard
 - Right (yellow) = Reactivity
 - Bottom (white) = Specific Hazard
- Remarks Fire Hazard:
 - 73 °F = 22.7°C
 - 100 °F = 37.7°C
 - 200 °F = 93.3°C



The **flash point** of a flammable liquid is the lowest temperature at which the surface of the liquid emits sufficient vapour to be ignited by a small flame.

Bottle label

A label on a bottle contains the following information:

- Name of the substance (usually in different languages).
- The purity:
 - Technical quality.
 - Pro-analyse
 - MOS quality
 - etc..
- The amount.
- The supplier.
- The hazard symbols.



Possible harm of chemicals to your body

Direct reaction:

- Corrosive : Attacks the body.
- Heat or burning effect : Causes burning wounds on the skin or eyes.
- Cryogenic burns : Freezing of the skin by liquid N₂ (at -196°C).

Delayed reaction in the body:

- Varies from irritation to an allergic reaction.
- Long term build-up effects (e.g. pollution by As, Hg or Pb).

Possible implications:

- | | |
|--|--|
| ▪ Carcinogenic (causes cancer) | ▪ Genotoxic (causes mutations), |
| ▪ Teratogenic (effects embryo growth) | ▪ Cytotoxic (attacks specific cells) |
| ▪ Reproductive toxic effect (influences fertility) | ▪ Neurotoxic (influences nervous system) |

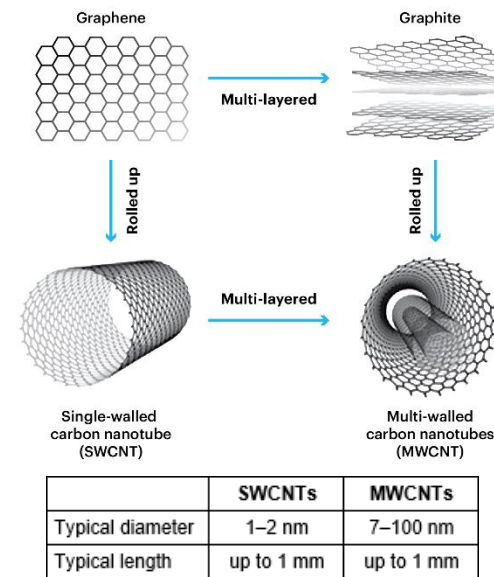
Possible harm caused by Nano-particles

Nano-particles are particles that have at least 1 dimension between 1-100 nm.

Not a lot of information is available about the influence of Nano-particles on peoples health. Always assume the **worst!** The knowledge about the harm that Nano-particles can cause to the human body is slowly increasing. Due to the large surface area they are highly reactive.

Possible health hazards:

- Absorption deep into the lungs.
- Migration through the skin / alveoli (lungs).
- Delayed reaction.



Ways of a chemical to enter your body

The hazardous effect of a chemical starts when it is in contact with or inside your body.

- A chemical can exist as a solid, liquid or gas.
- The danger of the intoxication depends on the chemical.

A chemical can enter the body by:

- Inhalation of a vapour when breathing. (Note that some chemicals have no smell.)
- Injection through an existing open wound.
- Through a wound that is caused by breaking your glassware during the experiment.
- Injection by the needle of a syringe.
- Absorption or penetration through the intact skin or the alveoli in the lungs.
- Via your mouth, stomach (if swallowed), ears and eyes.

Personal protection when working with chemicals

- Everyone needs a **buddy** to work with chemicals or liquid Nitrogen.
- Wear **safety glasses**. (If you already wear normal glasses, then over your normal glasses), or a **protective face shield** (also when you are looking over someone's shoulder).
- Wear **neoprene gloves** over the white nitrile cleanroom gloves.
 - Check if the Neoprene gloves are in good condition.
 - If the condition is not OK then throw them away and get new ones.
 - Rinse the neoprene gloves with DI-water when your ready.
- Use an apron to protect your body when you are pouring chemicals.
- Work in a fume hood or wet bench.



Rules for working with chemicals

- Read the MSDS of used chemicals before you start.
- Use your Personal Protective Equipment (PPE).
- It is mandatory to have a **buddy**.
- Always work in a lower part of **fume hood** or **wet bench**.
- Transport chemicals in a proper way to the working area:
 - Use a rubber bottle carrier for 1 bottle or the special trolley for transportation of multiple bottles.



Rubber bucket to transport 1 bottle.



Trolley for transportation of multiple bottles.

Report every possible hazardous situation to the EKL Safety Officer.

Not a good way to transport chemicals



Personal Protection Equipment (PPE) against chemicals



Transporting a bottle with a bottle carrier

A safer and easier way to transport chemicals is using a bottle carrier.

Inform your buddy about the chemical you are going to transport.

When you are going to transport a bottle with bottle carrier, you must:

- Wear protective clothing.
- Open the chemicals storage cabinet door with the nitrile gloves on.
- Put the neoprene glove on.
- Take the bottle with both hands out of the cabinet.
- Put it in the bottle carrier.
- Remove both neoprene gloves and close the storage cabinet door with the nitrile glove.
- Take the bottle carrier and go to the location where you will use the chemical.

Do not touch anything with the neoprene gloves, except the bottle.

Transporting a bottle with a trolley.

Another easy way to transport chemicals is using a trolley.

Inform your buddy about the chemical you are going to transport.

When you are going to transport a bottle with a trolley, you must:

- Wear protective clothing.
- Open the chemicals storage cabinet door with the nitrile gloves on.
- Put the neoprene glove on.
- Take the bottle with both hands out of the cabinet.
- Put it in the trolley.
- Remove both neoprene gloves and close the storage cabinet door with the nitrile glove.
- Roll the trolley to the location where you will use the chemical.

Do not touch anything with the neoprene gloves, except the bottle.

Restrictions regarding chemicals

Because of safety regulations it is **strictly forbidden** to:

- Order chemicals without approval.
- Bring your own chemicals without approval.
- Store chemicals in office areas.
- Create a mix of chemicals that has not been reviewed by your mentor, and is not approved by the EKL Safety Officer.

Due to safety rules it's mandatory to wear proper clothing under the CR-suit:

- No skirts/shorts.
- No open shoes.

Introducing new chemicals into the EKL labs

Procedure for using new chemicals in EKL:

- Read the MSDS.
- Consult your supervisor about the new chemical and your experimental setup.
- Contact the **Wet Bench team** by sending a mail to [Sara Ottoboni](#), with a cc to [Robert Verhoeven](#) and [Anastasiia Kotliarchuk](#). Ask for permission to use the chemical in the EKL cleanroom and attach the MSDS.
- After approval is granted, contact Robert and Anastasiia and supply the following information:
 - Name of the chemical.
 - Name of the manufacturer.
 - The cost, the amount and the quality (purity).
 - If requested by Robert, then supply the Baan code (financial code).
 - Mention in which lab you plan to use the new chemical.
 - Robert or Anastasiia will order it and provide you the information in which chemical storage cupboard you can find it.They will also make the MSDS available on the front of the cupboard.

Importance of good communication

We had a fire in a fume hood that could have been prevented by better communication causing less misunderstanding.

What went wrong?

- The user did not have the proper training.
- The experiment explanation by the user was misunderstood by the safety officer.
- The communication about the experimental setup was incomplete and wrongly interpreted.
- The chemicals were ordered by a staff member without approval from the safety officer.
- There was no buddy present during the mixing of the chemicals.
- During the process execution the user was not present.
- Mistakes made:
 - Heating up of an organic liquid directly on a hot plate.
 - Leaving the process un-attended.

Handling errors after discovering the fire:

- The person who discovered the fire called for assistance from a staff member instead of pushing the fire alarm button.
- The staff member pressed the wrong button (calamity button instead of the fire alarm button).
- The staff member filmed the fire on his mobile phone before extinguishing the fire.

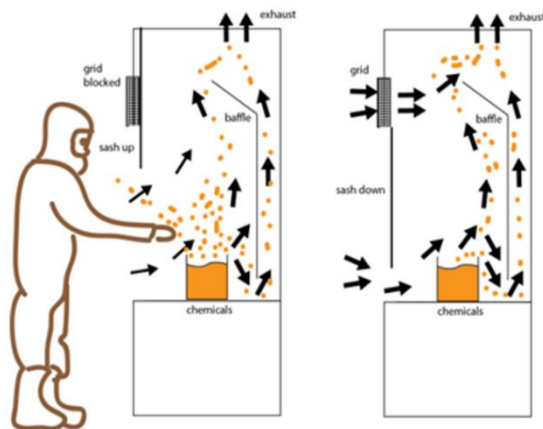
[Video: Fire in Fume hood.](#)

Work area

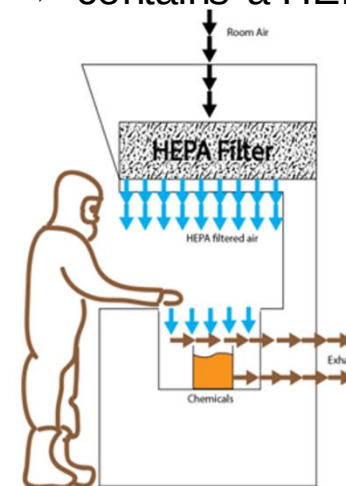
- A safe environment is required to work with chemicals, so that fumes will not be inhaled, and accidents can be avoided while using Personal Protective Equipment (PPE).
- Always use the lowest possible area of a wet bench or the sink of a fume hood while pouring chemicals.
- Always keep your working place clean.

EKL provides 2 types of locations:

① a fume hood ⇒ does not contain a HEPA filter



② a wet bench ⇒ contains a HEPA filter



- EKL has 2 wet bench lines:
- | | |
|-----------|--|
| CR 100 | ⇒ Mixing and pouring of chemicals is not allowed. |
| CR 10.000 | ⇒ Mixing and pouring of chemicals is allowed. |

Rules for working in a fume hood or wet bench

- A buddy is mandatory when you work with chemicals.
- Inform your buddy about the chemicals you are going to use.
- The buddy must be in the same room when you transport, handle or mix the chemicals.
- Use the personal protection equipment (PPE).
- Only one person is allowed work at the same time in a fume hood or wet bench.
- It is only allowed to create approved mixtures. **Do not create dangerous combinations.**
- Never sit on a chair while handling chemicals.
- Do not block the airflow at the back side of the fume hood.
- You need to know how to correctly open a chemical bottle.
- When the solution is made: Clean the outside of the bottle and put it back in the storage cabinet.
- Place a note on the fume hood sash with your name, telephone number and chemical mixture composition.
In case of an incident or accident the ERT then knows which chemicals are used.
- It is not allowed to store chemicals in a fume hood or wet bench without approval from the wet bench team and without a label.
- Always use the lowest possible area of a wet bench or the recessed surface of a fume hood while pouring chemicals. Do not work on the edge.
- **Never stick your head in a fume hood or wet bench.**

Wrong example of working with chemicals



Wrong example of working in a fume hood



How to open a chemical bottle

A bottle can have over-pressure that will release a vapour when it is opened.

Procedure to open a bottle:

- Inform your buddy about the chemicals that you are going to use.
- Your buddy must be in the same area.
- Use the PPE.
- Put the bottle on the recessed surface of the fume hood or on the top part of the wet bench in CR 10000.
- Push the bottle down with one hand to fix it and unlock the cap with the other hand.
You can hear if a vapour is released.
- When the cap is released, place the hand from top of the bottle to the side. Continue unscrewing the cap.
- Put the cap upside down on a higher clean surface of the fume hood.
- Pour the chemical in the glassware;
 - Beware that no chemical drops damage / react with the label on the bottle.
 - Always tilt the bottle in the direction of the non-labelled part when pouring.
- When finished pouring, place the cap back on the bottle and close it well.

How to clean a bottle after use

Before placing a bottle back in the storage cabinet, it needs proper cleaning. The bottle can be contaminated with drops of chemicals after pouring.

How to clean a bottle after use:

- You are still wearing the PPE.
- The buddy is still in the same area.
- Take a blue tissue and moisten it with DI-water.
- Be sure that the cap is tightened well.
- Take a DI-water pistol and spray the DI-water on the bottle while turning the bottle clockwise.
- Leave the bottle resting on the surface.
- Also softly spray upwards to rinse under the cap. Only clean the bottle and not the surrounding area.
- Wipe the bottle “dry” with the moistened bleu tissue. Wipe from the top to the bottom.
- Tilt the bottle gently and dry the bottom of the bottle.
- When ready, place the bottle on a dry clean higher area.

Wrong method of filling the measurement cylinder



Correct method: Use a small beaker



What to do if chemicals are spilled in a fume hood

A fume hood and a wet bench are specifically designed for pouring and mixing chemicals.

Cleaning of spilled chemicals in the fume hood/wet bench:

- In case of an **Acid** or a **Base** : Rinse with DI-water.
- In case of **Organic liquids** : Absorb with a blue tissue and dispose it in a special waste bin with an exhaust hood above it.

Tips while working with chemicals

While working with chemicals it is important that:

- Always protect yourself with the available PPE.
- Check if the neoprene gloves are in a good condition. If not OK then replace them.
- Be aware that your gloves can be contaminated with chemicals.
- Do not touch your body, clothing, doorknobs, phones, tablets, cleanroom notebooks, pens, equipment, etc. with the neoprene gloves. You can contaminate all the things you touch.
- Replace your nitrile gloves if they are possibly contaminated.
- Regularly clean and dry the neoprene gloves during the work, especially if you suspect a chemical contamination. Use DI-Water and a blue tissue.
- Clean all bottles before you place them back in the storage cabinet.
- When you are finished, wash and dry your neoprene gloves before taking them off.

Stay concentrated on what you are doing and keep your work area clean and dry.

Checks before starting an experiment

Checks outside the Labs:

- Do you have a process run number?
- Is your flowchart checked and approved?
- Did you finish your RISK assessment and is it approved?

Check inside the labs:

- Check if a sufficiently amount of the chemicals that you want to use is available in the cleanroom storage cabinets.
- Check if all the glassware that you need is available.
- Know how to store your chemicals and/or how to dispose them.
- Check if there are enough CR tissues.

If some of the above items are missing or are not available in the needed amount:

Contact the Wet Bench Team.

Additional (safety) information

- The chemicals in the bottles are “pure”.
Do not contaminate them by pouring residues back into them.
- Do not use damaged glassware.
Put the broken glass only in the waste bin for broken-glass.
- For heating up chemicals: Use a heater with a stirrer module and check if the thermocouple (needed for temperature control) is connected and submerged deep enough in the liquid.
- Use Pyrex or Duran glassware, as this is suitable for thermal processing.
- Use an Au Bain-Marie setup for heating up of organic liquids.
- Never try to catch falling glassware. Especially if it contains chemicals.

Rules for mixing chemicals

In most cases when chemicals are mixed, there will be a chemical reaction.

Dilution of an acid or a base with water gives an exothermic reaction. The solution and glassware are heating up.

While mixing chemicals:

- Always fill up the beaker with DI-water first before adding the other chemical under continuous mixing.
 - When mixing more than one chemical: Start with adding the chemical with lowest concentration first to the water.
 - Please note that a liquid with a higher density than water sinks to be on the bottom of the mixture. This means that it is necessary to stir well. E.g. the density of $\text{H}_3\text{PO}_4 = 1.7\text{kg/l}$.
- Use a mechanical stirrer or stir manual during the mixing.
- Always use a stirrer while heating up chemicals and keep stirring during the whole experiment.
- Because most solutions have a significant temperature rise during the mixing, it is advised to wait for the solution to reach the correct experiment temperature before starting your process. E.g. etch rates can be significantly different (usually 10°C more results in a 2 times larger etch rate).

Haste makes waste

Rushing substantially increases the change for incidents or accidents.

When you work with chemicals it is advisable to make a time schedule for your experiment that includes the time that is needed for:

- Finding a buddy.
- Finding the right size of neoprene gloves.
- Collecting the needed chemicals, glassware, enough tissues and nitrile gloves.
- Mixing of the chemicals and cleaning the bottles.
- Cooling down or heating up of the mixture.
- The experiment itself (including rinsing with DI-water and measuring of the result).
- Cleaning up the mixture and the work area.

Before performing a first time experiment on your process wafers, it is advised to use a test wafer first. During this test experiment you learn what is the best way to perform the experiment (e.g. handling of the used equipment), and to determine etch/deposition rates or other process parameters.

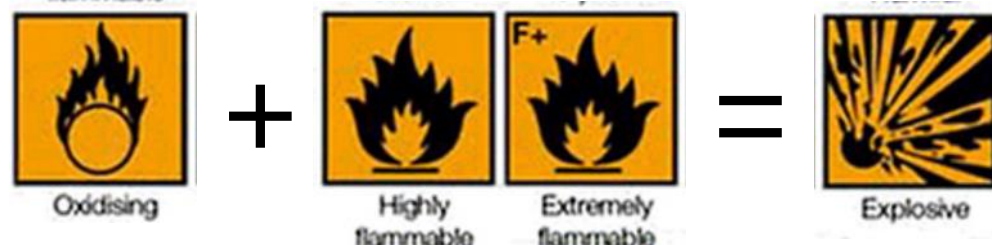
Dangerous combinations

The reaction between certain chemicals can be so violent that it is not allowed to mix them in any of the EKL laboratories. **These chemicals are called incompatibles.**

There is a long list of incompatible chemicals. A few categories are:

- Oxidizing – reducing agents.
 - Peroxides/HNO₃ – Organics. [Video: Nitric acid and Ethanol.](#)

If we look at the labels:



- Acids – Bases
 - HCl - KOH

Chemical storage/disposal cabinets

There are 3 different cabinets:

- **Acid/base cabinet:**

A ventilated cabinet with compartments: Acids are separated from the Bases.

- **Fire safe cabinet:**

A ventilated cabinet for all organic chemicals.

- **Disposal cabinet:**

A ventilated cabinet for all waste containers.

The cabinets below the fume hoods and wet benches are dedicated for one kind of chemical only.

They are all ventilated cabinets.

Do not place empty bottles back in any kind of storage cabinet. Place them on the correct spot for disposal.

Storage of (re)useable chemicals

It is possible to store your (preapproved) created mixtures.

Keep in mind that some mixtures (e.g. H_2O_2 containing mixtures) will degrade with time.

How to store your mixtures:

- Ask the wet bench team ([Robert Verhoeven](#) or [Anastasiia Kotliarchuk](#)) for the correct bottles.
(Strong oxidisers in glass, HF in plastic/Teflon).
- Also ask where to store the bottle(s).
- Add a label to the bottle(s) and mark them with waterproof ink with:
 - Your name and phone number.
 - The date when the solution is created.
 - All names of the contained chemicals and concentrations.
- After finishing the experiments, you remove the bottle(s) that you have in the storage.
 - Dispose your chemicals according to the correct procedures
 - Clean the bottle(s) with DI-water and dispose them in the correct waste bin.

Disposal rules for chemicals

Chemical classification:

■ Inorganic

- Group I: Acid and neutral mixtures
- Group II: Base and neutral mixtures

Disposal

Water jet pump
Water jet pump

Container label



■ Organic

- Group III: Non Halogens containing organics
- Group IV: Halogen containing chemicals
(F, Cl, Br, I)

Container III
Container IV



■ Special waste

- Group V: Special waste
- Group VI: Special waste with exceptional risk

Container V
Container VI



When the etchant of group I and II contains a lot of environmental dangerous materials:

Then the etchant must be disposed in a container from cat. I or II. Attach a label on the container and write down the contents.

Finishing up after the experiments

**You need a buddy inside the area where you work, because:
You are handling chemicals.**

What to do?

- Dispose the chemicals according the disposal rules.
(See slide: [Disposal rules for chemicals](#)).
- Rinse the used glassware 3 times with DI-water and put it upside down in the tray for dirty glassware.
(Select the correct one: for not-contaminated or contaminated glassware).
- Rinse the recessed surface of the fume hood or the used wet bench surface with DI-water.
- Clean the top surface of the fume hood with a moistened blue tissue and dry it with a dry blue tissue.
- Clean all the touched water valve buttons, the N₂ and DI-water guns and the sash handles with a new moistened blue tissue .
- Leave the fume hood in a proper clean state.

Report any problems/safety issues to the EOD, the wet bench team or the safety officer [Sara Ottoboni](#).

When leaving the cleanroom: You ALWAYS wash your hands!



NaOH in water <https://www.youtube.com/watch?v=E6Dlf47sLuU>

NaOH HCl <https://www.youtube.com/watch?v=rL3k8JJhci4>

Chicken leg in Piranha https://www.youtube.com/watch?v=CTVd_WxblGI